



Dirty Dog Labs Network Analysis

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July 1/2024

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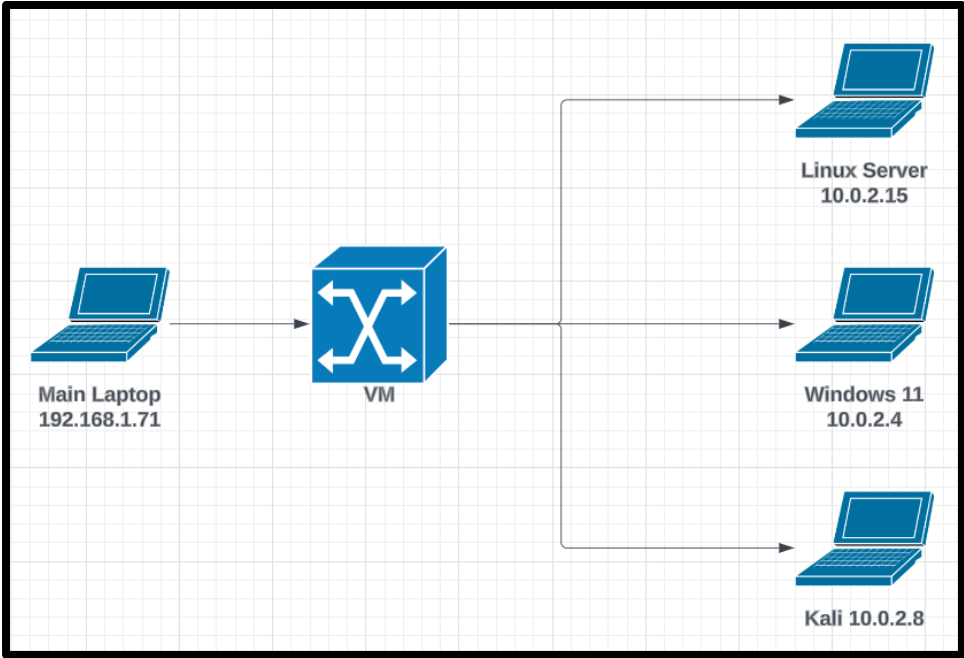
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Current Network Topology



Executive Summary

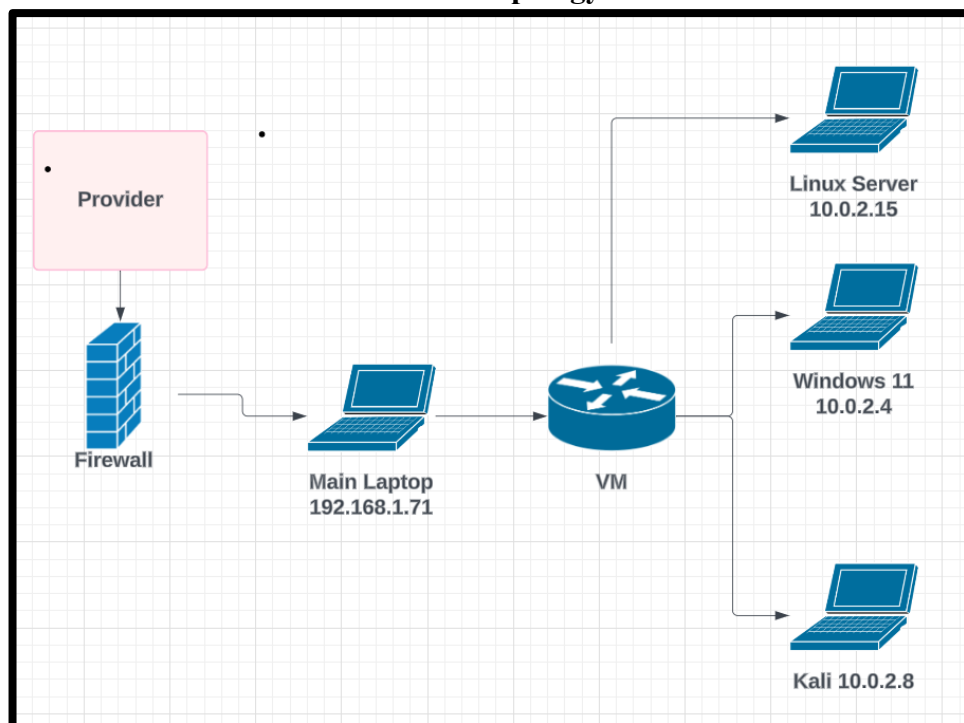
Security assessment of 192.168.1.71 network identified three devices: a Kali Linux machine, Windows 11 machine and a Linux server. Overall this network is stable, with limited minor security vulnerabilities. The initial network topography shows how the network is connected, mainly the network is dependent on VM aka virtual machine to manage resources of the host main computer.

Security Recommendations:

- **Firewall:** Implement a firewall between the main computer and internet provider

- **VLAN segmentation:** Separate networks for better protection
- **Patch management:** automate antivirus and software updates
- **Strong authentication:** enforce strong password policies, and multi factor authorization when able.
- **Secure network services:** review configurations of networks
- **Restrict network access:** limit access to critical data by implementing access control
- **Network traffic monitoring:** plan and implement network monitoring to ensure quick responds to suspicious activity

Future topology



Network Device Information

Main Laptop: The device is identified as a virtual box and internal DNS server made by Oraclebox. For ideal clarity this device will be excluded from the following report since it does not pose security concerns.

Windows(ref. Image 16-21)

- Hostname: **WINDOWS11 - DESKT**
- IP Address: **10.0.2.4 / 24**
 - **192.168.1.77**
- MAC: **08-00-27-CB-20-4A**
- OS: **Windows 11 23H2**

- Version: **10.0.22631**
- Open ports and associated services:
 - **Http (80/tcp)**
 - **Msrpc (135/tcp)**
 - **Netbios-ssn (139/tcp)**
 - **Microsoft-ds (445/tcp)**
- ARP Ping Scan time: **93.81 seconds**
- OSI Layer header & port #
 - **Src IP: 10.0.2.4 Src Port: 80**
 - **Dst IP: 10.0.2.8 Dst Port: 50768**

Kali (ref. Image 11-15)

- Hostname: **Kali**
- IP Address: **10.0.2.8 /24**
 - **192.168.1.41 (from Linux server and windows 11)**
- MAC: **08:00:27:1b:76:b0**
- OS: **Kali GNU/Linux rolling x86_64**
- Version: **6.8.11-amd64**
- Open ports: **Http (80/tcp)**
- ARP Ping Scan time: **N/A (Host**
- OSI Layer header & port #
 - **N/A (Host**

Linux Server (ref. Image 16-21)

- Hostname: **Linux Server**
- IP Address: **10.0.3.15 /24**
 - **192.168.1.76**
- MAC: **08:00:27:dd:d8:f8**
- OS: **Ubuntu 22.04.4**
- Kernel: **6.5.0-35-generic**
- Open ports:
 - **Http (80/tcp)**
- ARP Ping Scan time: **0.06 seconds**
- OSI Layer header & port #
 - **Src IP: 10.0.2.8 Src Port: 46376**
 - **Dst IP: 192.168.1.76 Dst Port: 80**

Information Collection Methodology

The network security assessment conducted to identify and verify all connected devices on the (Main laptop IP) network segment. Though the assessment nmap vulnerability scanner was running on the Kali Linux virtual machine. Which gathered Ip address, MAC addresses, Operating systems, open ports and response time details. Further, Wireshark, the network traffic analysis tool was used to examine data packet communications. The analysis filters in wireshark identified source and destination ip and ports. Finally, commands such as 'ip addr', 'neofetch', 'fastfetch', 'ifconfig', and arp were used in each machine terminal to verify information and configurations. This allowed for a multi-pronged approach to comprehensively assess the security of the network.

Conclusion

The network is well managed by the virtual machine but could benefit from firewalls between host and provider. Further the resources of the main laptop are limited so adding additional resources would prevent slow, lagging or complete crashing of the windows, kali and linux server.

References

Bohlooli, A. M. (2021, November 27). *How to install and use neofetch on windows*. MUO. <https://www.makeuseof.com/how-to-install-and-use-neofetch-on-windows/>

Bohlooli, A. M. (2022, March 17). *How to install scoop in windows*. MUO. <https://www.makeuseof.com/windows-install-scoop/>

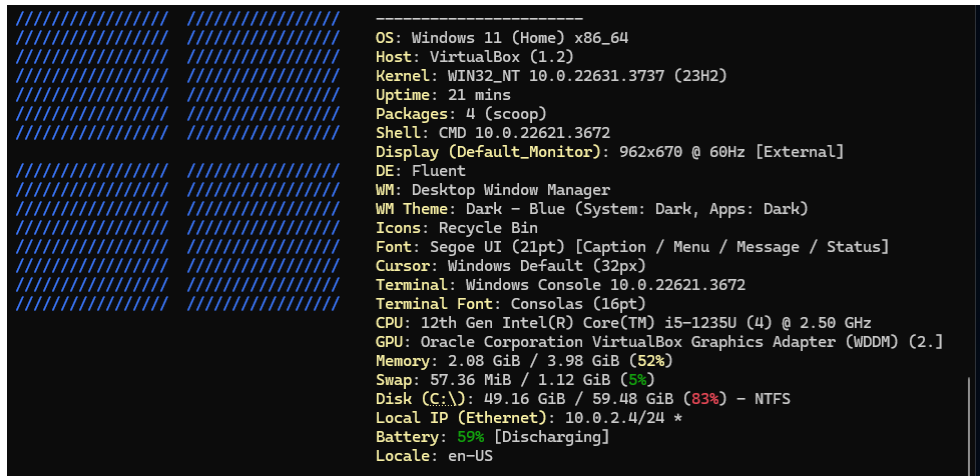
Citation generator. Scribbr. (2024, March 21). <https://www.scribbr.com/citation/generator/>

Dylananaraps. (n.d.). *How install neofetch win 11 · issue #2438 · Dylananaraps/Neofetch*. GitHub. <https://github.com/dylananaraps/neofetch/issues/2438>

Intelligent diagramming. Lucidchart. (n.d.). <https://www.lucidchart.com/pages/>

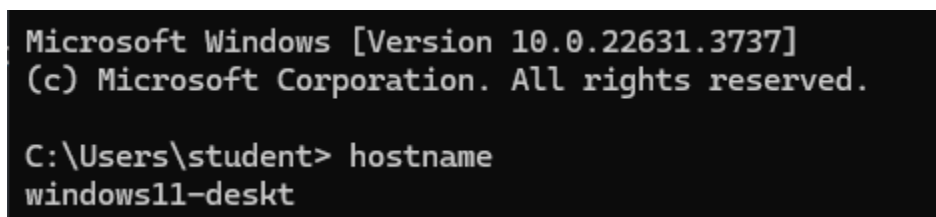
Screen Captures

Window images



```
OS: Windows 11 (Home) x86_64
Host: VirtualBox (1.2)
Kernel: WIN32_NT 10.0.22631.3737 (23H2)
Uptime: 21 mins
Packages: 4 (scoop)
Shell: CMD 10.0.22621.3672
Display (Default_Monitor): 962x670 @ 60Hz [External]
DE: Fluent
WM: Desktop Window Manager
WM Theme: Dark - Blue (System: Dark, Apps: Dark)
Icons: Recycle Bin
Font: Segoe UI (21pt) [Caption / Menu / Message / Status]
Cursor: Windows Default (32px)
Terminal: Windows Console 10.0.22621.3672
Terminal Font: Consolas (16pt)
CPU: 12th Gen Intel(R) Core(TM) i5-1235U (4) @ 2.50 GHz
GPU: Oracle Corporation VirtualBox Graphics Adapter (WDDM) (2.)
Memory: 2.08 GiB / 3.98 GiB (52%)
Swap: 57.36 MiB / 1.12 GiB (5%)
Disk (C:): 49.16 GiB / 59.48 GiB (83%) - NTFS
Local IP (Ethernet): 10.0.2.4/24 *
Battery: 59% [Discharging]
Locale: en-US
```

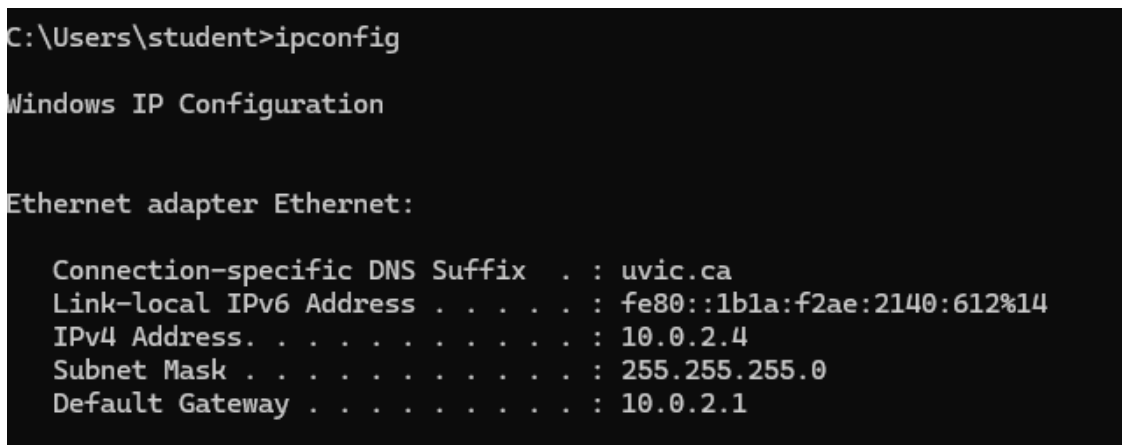
Figure 1: fastfetch of Windows (fastfetch is similar to neofetch)



```
Microsoft Windows [Version 10.0.22631.3737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student> hostname
windows11-deskt
```

Figure 2: confirm hostname of windows



```
C:\Users\student>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : uvic.ca
    Link-local IPv6 Address . . . . . : fe80::1b1a:f2ae:2140:612%14
    IPv4 Address. . . . . : 10.0.2.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1
```

Figure 3: windows ip configuration

```
Windows IP Configuration

Host Name . . . . . : windows11-deskt
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : uvic.ca

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : uvic.ca
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-CB-20-4A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1b1a:f2ae:2140:612%14(Preferred)
IPv4 Address. . . . . : 10.0.2.4(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, July 14, 2024 7:07:52 PM
Lease Expires . . . . . : Sunday, July 14, 2024 7:22:41 PM
Default Gateway . . . . . : 10.0.2.1
DHCP Server . . . . . : 10.0.2.3
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-94-FB-BD-08-00-27-CB-20-4A
DNS Servers . . . . . : 142.104.6.1
                        142.104.80.2
NetBIOS over Tcpip. . . . . : Enabled
```

Figure 4: Windows ip config command

```
C:\Users\student>arp -a

Interface: 192.168.1.77 --- 0xe
Internet Address      Physical Address      Type
192.168.1.41          8c-83-94-0d-ad-e0     dynamic
192.168.1.71          ac-5a-fc-bb-20-e7     dynamic
192.168.1.254         8c-83-94-9c-a5-79     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22           01-00-5e-00-00-16     static
224.0.0.251          01-00-5e-00-00-fb     static
224.0.0.252          01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Figure 5: arp command on Windows

```
C:\Users\student>nmap -Pn 192.168.1.77
Starting Nmap 7.94 ( https://nmap.org ) at 2024-07-03 01:28 Eastern Daylight Time
Nmap scan report for 192.168.1.77
Host is up (0.0054s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

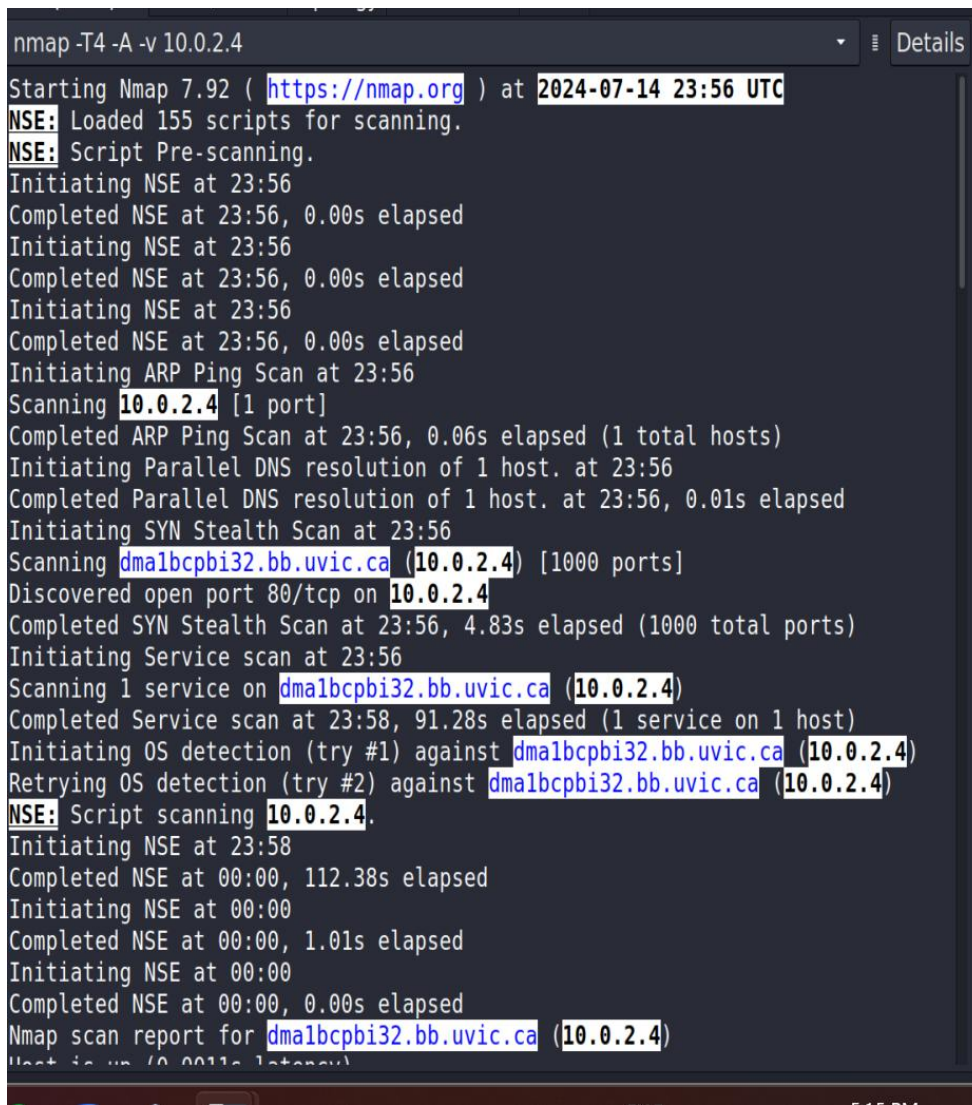
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds
```

Figure 6: windows nmap

```
C:\Users\student>systeminfo

Host Name:                WINDOWS11-DESKT
OS Name:                  Microsoft Windows 11 Home
OS Version:               10.0.22631 N/A Build 22631
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         Windows User
Registered Organization:   00326-10000-00000-AA746
Product ID:                00326-10000-00000-AA746
Original Install Date:     3/26/2024, 2:55:29 PM
System Boot Time:          7/1/2024, 6:59:23 PM
System Manufacturer:       innotek GmbH
System Model:              VirtualBox
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 154 Stepping 4 GenuineIntel ~2496 Mhz
BIOS Version:              innotek GmbH VirtualBox, 12/1/2006
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume2
System Locale:              en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC-05:00) Eastern Time (US & Canada)
Total Physical Memory:     4,080 MB
Available Physical Memory: 1,150 MB
Virtual Memory: Max Size:  5,488 MB
Virtual Memory: Available: 2,023 MB
Virtual Memory: In Use:    3,465 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\WINDOWS11-DESKT
Hotfix(s):                 6 Hotfix(s) Installed.
                           [01]: KB5037591
                           [02]: KB5027397
                           [03]: KB5036212
                           [04]: KB5039212
                           [05]: KB5035967
                           [06]: KB5037959
Network Card(s):           2 NIC(s) Installed.
                           [01]: Intel(R) PRO/1000 MT Desktop Adapter
                               Connection Name: Ethernet
                               DHCP Enabled:   Yes
                               DHCP Server:    192.168.1.254
                               IP address(es)
                               [01]: 192.168.1.77
                               [02]: fe80::1b1a:f2ae:2140:612
                               [03]: 2001:569:ba84:5f00:61c3:6fell:ef53:ce97
                               [04]: 2001:569:ba84:5f00:6225:2054:39c1:d96c
                           [02]: Intel(R) PRO/1000 MT Desktop Adapter
                               Connection Name: Ethernet 2
                               Status:         Hardware not present
Hyper-V Requirements:      A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

Figure 7: windows system info



```
nmap -T4 -A -v 10.0.2.4
Starting Nmap 7.92 ( https://nmap.org ) at 2024-07-14 23:56 UTC
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 23:56
Completed NSE at 23:56, 0.00s elapsed
Initiating NSE at 23:56
Completed NSE at 23:56, 0.00s elapsed
Initiating NSE at 23:56
Completed NSE at 23:56, 0.00s elapsed
Initiating ARP Ping Scan at 23:56
Scanning 10.0.2.4 [1 port]
Completed ARP Ping Scan at 23:56, 0.06s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 23:56
Completed Parallel DNS resolution of 1 host. at 23:56, 0.01s elapsed
Initiating SYN Stealth Scan at 23:56
Scanning dmalbcpbi32.bb.uvic.ca (10.0.2.4) [1000 ports]
Discovered open port 80/tcp on 10.0.2.4
Completed SYN Stealth Scan at 23:56, 4.83s elapsed (1000 total ports)
Initiating Service scan at 23:56
Scanning 1 service on dmalbcpbi32.bb.uvic.ca (10.0.2.4)
Completed Service scan at 23:58, 91.28s elapsed (1 service on 1 host)
Initiating OS detection (try #1) against dmalbcpbi32.bb.uvic.ca (10.0.2.4)
Retrying OS detection (try #2) against dmalbcpbi32.bb.uvic.ca (10.0.2.4)
NSE: Script scanning 10.0.2.4.
Initiating NSE at 23:58
Completed NSE at 00:00, 112.38s elapsed
Initiating NSE at 00:00
Completed NSE at 00:00, 1.01s elapsed
Initiating NSE at 00:00
Completed NSE at 00:00, 0.00s elapsed
Nmap scan report for dmalbcpbi32.bb.uvic.ca (10.0.2.4)
Host is up (0.001s latency)
```

Figure 8: zenmap nmap details for windows 1

```
Discovered open port 80/tcp on 10.0.2.4
Completed SYN Stealth Scan at 23:56, 4.83s elapsed (1000 total ports)
Initiating Service scan at 23:56
Scanning 1 service on dma1bcpbi32.bb.uvic.ca (10.0.2.4)
Completed Service scan at 23:58, 91.28s elapsed (1 service on 1 host)
Initiating OS detection (try #1) against dma1bcpbi32.bb.uvic.ca (10.0.2.4)
Retrying OS detection (try #2) against dma1bcpbi32.bb.uvic.ca (10.0.2.4)
NSE: Script scanning 10.0.2.4.
Initiating NSE at 23:58
Completed NSE at 00:00, 112.38s elapsed
Initiating NSE at 00:00
Completed NSE at 00:00, 1.01s elapsed
Initiating NSE at 00:00
Completed NSE at 00:00, 0.00s elapsed
Nmap scan report for dma1bcpbi32.bb.uvic.ca (10.0.2.4)
Host is up (0.0011s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      PRTG
|_http-server-header: PRTG
|_http-favicon: Unknown favicon MD5: 36B3EF286FA4BEFBB797A0966B456479
|_fingerprint-strings:
|_  FourOhFourRequest:
|_    HTTP/1.1 302 Moved Temporarily
|_    Connection: close
|_    Content-Type: text/html; charset=utf-8
|_    Content-Length: 54
|_    Date: Sun, 14 Jul 2024 23:56:46 GMT
|_    Expires: 0
|_    Cache-Control: no-cache
|_    X-Content-Type-Options: nosniff
|_    X-XSS-Protection: 1; mode=block
```

Figure 9: zenmap nmap for windows 2

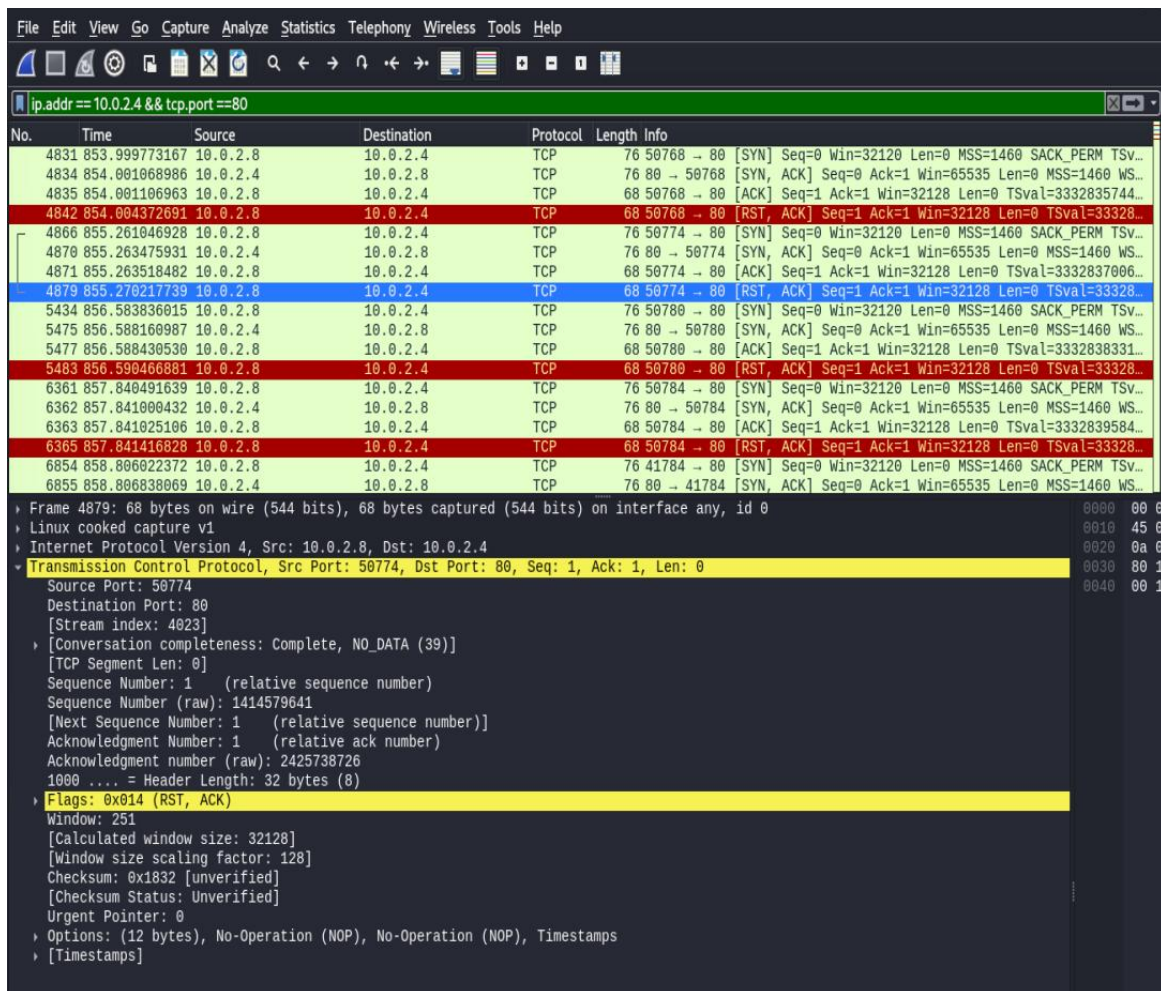


Figure 10: Wireshark for windows, source and destination ports

Kali images

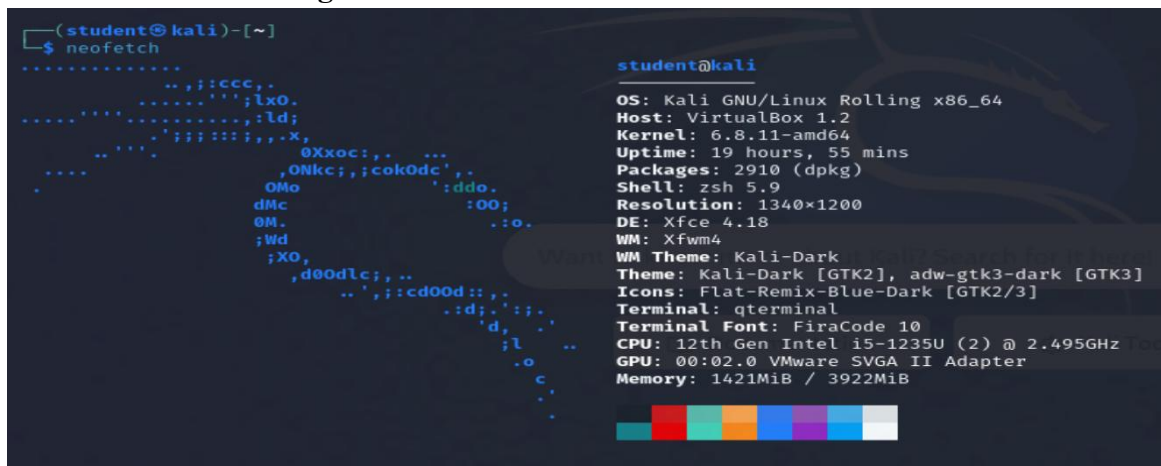


Figure 11: neofetch of Kali

```
student@kali: ~  
File Actions Edit View Help Pure Analyze Statistics Telephony Wireless Tools Help  
(student@kali)~  
$ ifconfig  
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255  
    ether 02:42:c3:c3:12:f4 txqueuelen 0 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.8 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fe80::a00:27ff:fe1b:76b0 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:1b:76:b0 txqueuelen 1000 (Ethernet)  
    RX packets 1728 bytes 122814 (119.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 4437 bytes 335776 (327.9 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 48 bytes 2480 (2.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 48 bytes 2480 (2.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(student@kali)~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:1b:76:b0 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.8/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 487sec preferred_lft 487sec  
    inet6 fe80::a00:27ff:fe1b:76b0/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
    link/ether 02:42:c3:c3:12:f4 brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
  
(student@kali)~  
$  
(student@kali)~
```

Figure 12: command line screenshot of kali ifconfig and ip addr

```
(student@kali)~  
$ sudo lsb_release -a  
[sudo] password for student:  
No LSB modules are available.  
Distributor ID: Kali  
Description:    Kali GNU/Linux Rolling  
Release:        2024.2  
Codename:       kali-rolling
```

Figure 13: Kali release for OS details and hostname

```
(student@kali)~  
$ arp -a  
? (10.0.2.3) at 08:00:27:02:7b:79 [ether] on eth0  
? (10.0.2.3) at 08:00:27:02:7b:79 [ether] on eth0  
(student@kali)~
```

Figure 14: Kali arp screenshot of variations found

```
File Actions Edit View Help
(student@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1b:76:b0 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.8/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 600sec preferred_lft 600sec
    inet6 fe80::a00:27ff:fe1b:76b0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:c2:c3:12:f4 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(student@kali)-[~]
$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:c2:c3:12:f4 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.8 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe1b:76b0 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:1b:76:b0 txqueuelen 1000 (Ethernet)
    RX packets 4033 bytes 276816 (270.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11291 bytes 851313 (831.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 48 bytes 2480 (2.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 48 bytes 2480 (2.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(student@kali)-[~]
$ nmap -Pn 192.168.1.41
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-07-03 02:27 EDT
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 37.80% done; ETC: 02:27 (0:00:13 remaining)
Nmap scan report for 192.168.1.41
Host is up (0.0055s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 13.52 seconds

(student@kali)-[~]
$ arp -a
? (10.0.2.1) at 52:54:00:12:35:00 [ether] on eth0
? (10.0.2.3) at 08:00:27:02:7b:79 [ether] on eth0

(student@kali)-[~]
$ neofetch
```

Figure 15: Kali overview

Linux server images

```
student@linux-server:~$ neofetch

      .-/+oosssso+/-.
      `:+ssssssssssssssssss+:`
      -+ssssssssssssssssssyyssss+-
      .osssssssssssssssssdMMMMyssso.
      /ssssssssssshdmmNNmyNMMMMhsssss/
      +ssssssssshmycMMMMMMNddddyssssss+
      /ssssssssshNMMMyhhyyyhNMMMMhssssss+/
      .ssssssssdMMMNhssssssssshNMMMdssssss.
      +ssssshhhyNMMNyssssssssssyNMMMyssssss+
      ossyNMMMNyMMhssssssssssshmmhssssssso
      ossyNMMMNyMMhssssssssssshmmhssssssso
      +ssssshhhyNMMNyssssssssssyNMMMyssssss+
      .ssssssssdMMMNhssssssssshNMMMdssssss.
      /ssssssssshNMMMyhhyyyhdNMMMMhssssss+/
      +ssssssssshmycMMMMMMNddddyssssss+
      /ssssssssssshdmmNNmyNMMMMhssssss/
      .osssssssssssssssssdMMMMyssso.
      -+ssssssssssssssssyyssss+-
      `:+ssssssssssssssssss+:`
      .-/+oosssso+/-.

student@linux-server
-----
OS: Ubuntu 22.04.4 LTS x86_64
Host: VirtualBox 1.2
Kernel: 6.5.0-35-generic
Uptime: 17 hours, 47 mins
Packages: 1797 (dpkg), 10 (snap)
Shell: bash 5.1.16
Resolution: 1280x800
DE: GNOME 42.9
WM: Mutter
WM Theme: Adwaita
Theme: Yaru [GTK2/3]
Icons: Yaru [GTK2/3]
Terminal: gnome-terminal
CPU: 12th Gen Intel i5-1235U (4) @ 2
GPU: 00:02.0 VMware SVGA II Adapter
Memory: 1585MiB / 3906MiB
```

Figure 16: neofetch of linux server

```
student@linux-server:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.76 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::82e1:f9bc:62f2:f172 prefixlen 64 scopeid 0x20<link>
    inet6 2001:569:ba84:5f00:c066:e6a9:2879:17e4 prefixlen 64 scopeid 0x0<global>
    inet6 2001:569:ba84:5f00:b2a8:b678:3454:9d62 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:dd:d8:f8 txqueuelen 1000 (Ethernet)
    RX packets 255796 bytes 284962360 (284.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29502 bytes 2245352 (2.2 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::da66:3540:5b3b:66f5 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:89:ae:2d txqueuelen 1000 (Ethernet)
    RX packets 541 bytes 87692 (87.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 708 bytes 66356 (66.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 497 bytes 42905 (42.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 497 bytes 42905 (42.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 17: Linux Server ifconfig command

```
try: sudo apt install <deb name>
student@linux-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:dd:d8:f8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.76/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 18070sec preferred_lft 18070sec
    inet6 2001:569:ba84:5f00:b2a8:b678:3454:9d62/64 scope global temporary dynamic
        valid_lft 14577sec preferred_lft 14277sec
    inet6 2001:569:ba84:5f00:c066:e6a9:2879:17e4/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 14577sec preferred_lft 14277sec
    inet6 fe80::82e1:f9bc:62f2:f172/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:89:ae:2d brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8
        valid_lft 77486sec preferred_lft 77486sec
    inet6 fe80::da66:3540:5b3b:66f5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
student@linux-server:~$
```

Figure 18: Linux Server ip addr command

```
student@linux-server:~$ arp -a
Telus (192.168.1.254) at 8c:83:94:9c:a5:79 [ether] on enp0s3
? (192.168.1.41) at 8c:83:94:0d:ad:e0 [ether] on enp0s3
? (192.168.1.65) at c8:12:0b:58:de:be [ether] on enp0s3
? (192.168.1.71) at ac:5a:fc:bb:20:e7 [ether] on enp0s3
_gateway (10.0.3.2) at 52:54:00:12:35:02 [ether] on enp0s8
student@linux-server:~$ nmap -Pn 192.168.1.76
Starting Nmap 7.95 ( https://nmap.org ) at 2024-07-03 01:35 EDT
Nmap scan report for linux-server (192.168.1.76)
Host is up (0.00020s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
student@linux-server:~$
```

Figure 19:Linux server arp and nmap commands screenshot

```
nmap -T4 -A -v 10.0.2.15
Starting Nmap 7.92 ( https://nmap.org ) at 2024-07-14 23:51 UTC
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Initiating ARP Ping Scan at 23:51
Scanning 10.0.2.15 [1 port]
Completed ARP Ping Scan at 23:51, 1.45s elapsed (1 total hosts)
Nmap scan report for 10.0.2.15 [host down]
NSE: Script Post-scanning.
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Initiating NSE at 23:51
Completed NSE at 23:51, 0.00s elapsed
Read data files from: /usr/local/bin/./share/nmap
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.05 seconds
Raw packets sent: 2 (56B) | Rcvd: 0 (0B)
```

Figure 20: Zenmap nmap for Linux Server

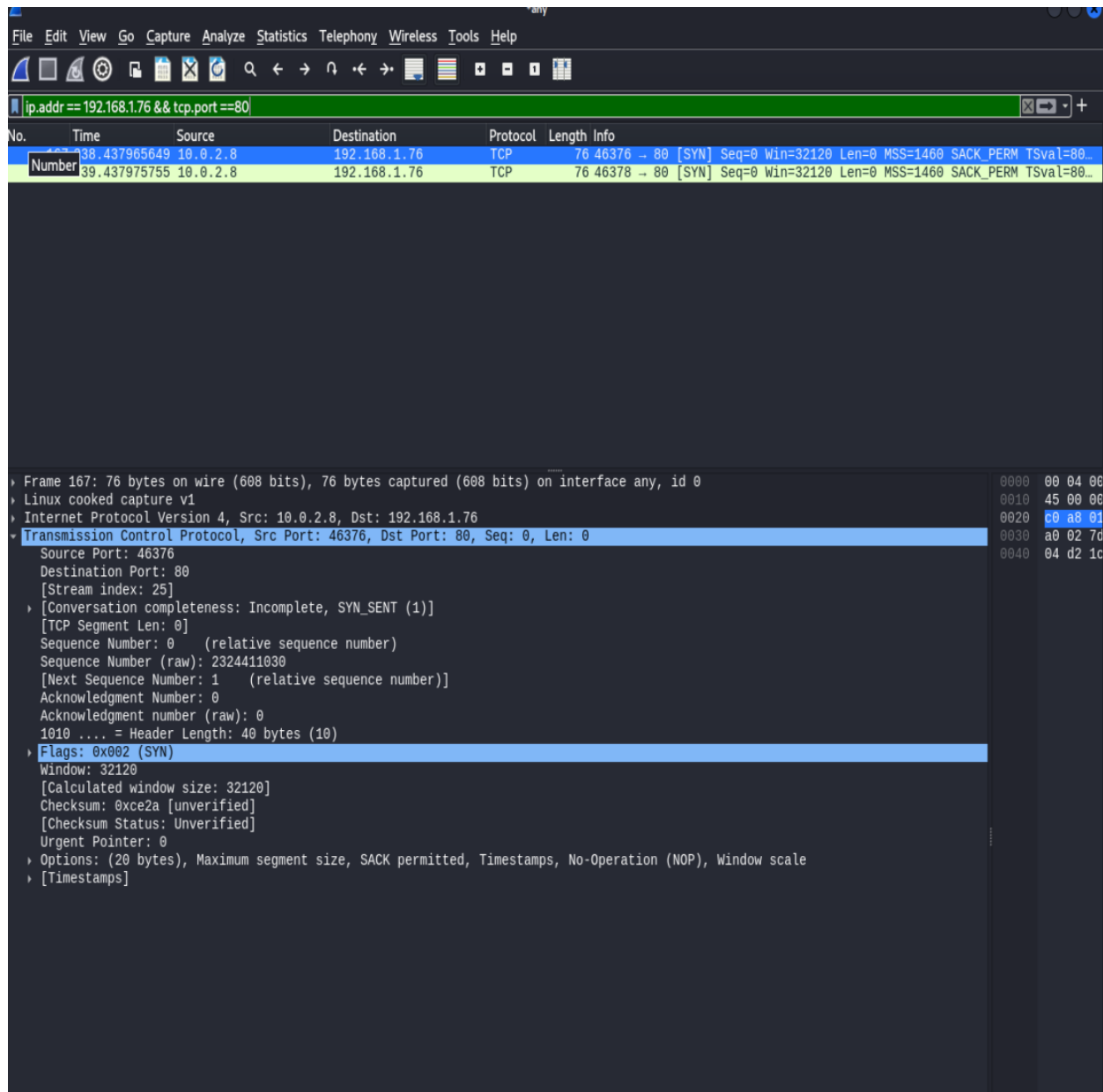


Figure 21: Wireshark for Linux Server